

Calibrated DistilBERT for Reliable Extractive QA

SQuAD 2.0 task adaptation + held-out no-answer calibration

CS224N default final project | Independent course study

Problem & research question

Extractive QA models can return a convincing span even when a passage does not contain an answer. SQuAD 2.0 makes this failure explicit with adversarial unanswerable questions.

Question: How much do SQuAD 2.0 fine-tuning and a separately calibrated no-answer threshold improve a compact model trained only on SQuAD 1.1?

Controlled design

1. Same 66.4M-parameter DistilBERT architecture.
2. Measure task adaptation separately from threshold calibration.
3. Never use final evaluation labels to select the threshold.

Data & experimental setup

Train: 12,000 examples / 13,322 windowed features
Calibration: 1,500 examples
Final evaluation: 1,500 examples (743 answerable, 757 unanswerable)
Input: 256 wordpieces, 64-token stride
Training: 2 epochs, AdamW, lr 3e-5, effective batch 24, bfloat16
Hardware: RTX 5060 Laptop GPU, 8 GB
Training time: 274 seconds

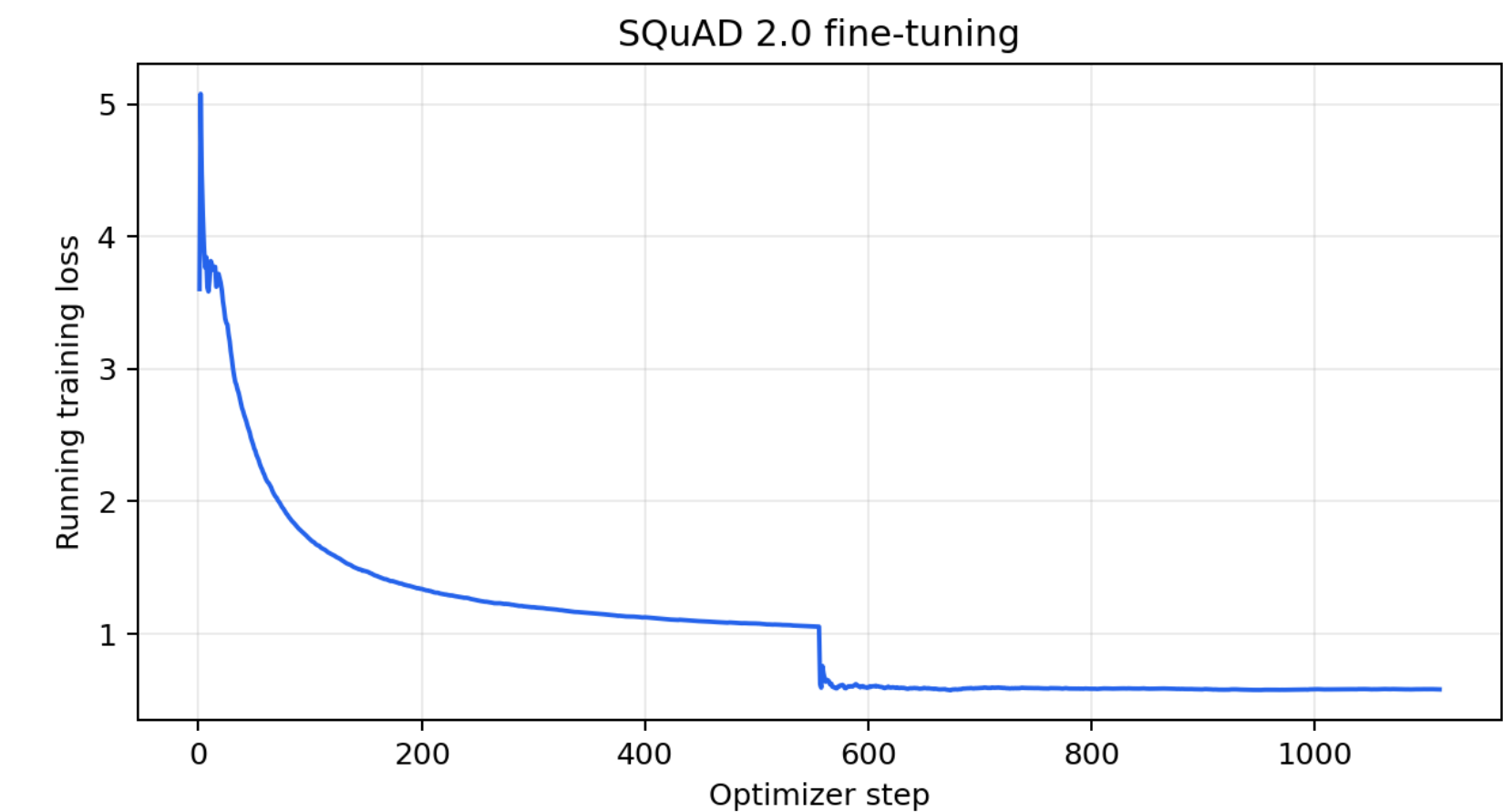
Metrics: normalized Exact Match, token F1, answerable-only F1, and unanswerable accuracy.

Method

Question and context are encoded jointly. A QA head predicts start and end logits for every token. Candidate spans are pooled across overlapping context windows.

span score = $I_s(i) + I_e(j)$
null score = $I_s([CLS]) + I_e([CLS])$
 $d = \text{null score} - \text{best span score}$
abstain when $d > \tau$

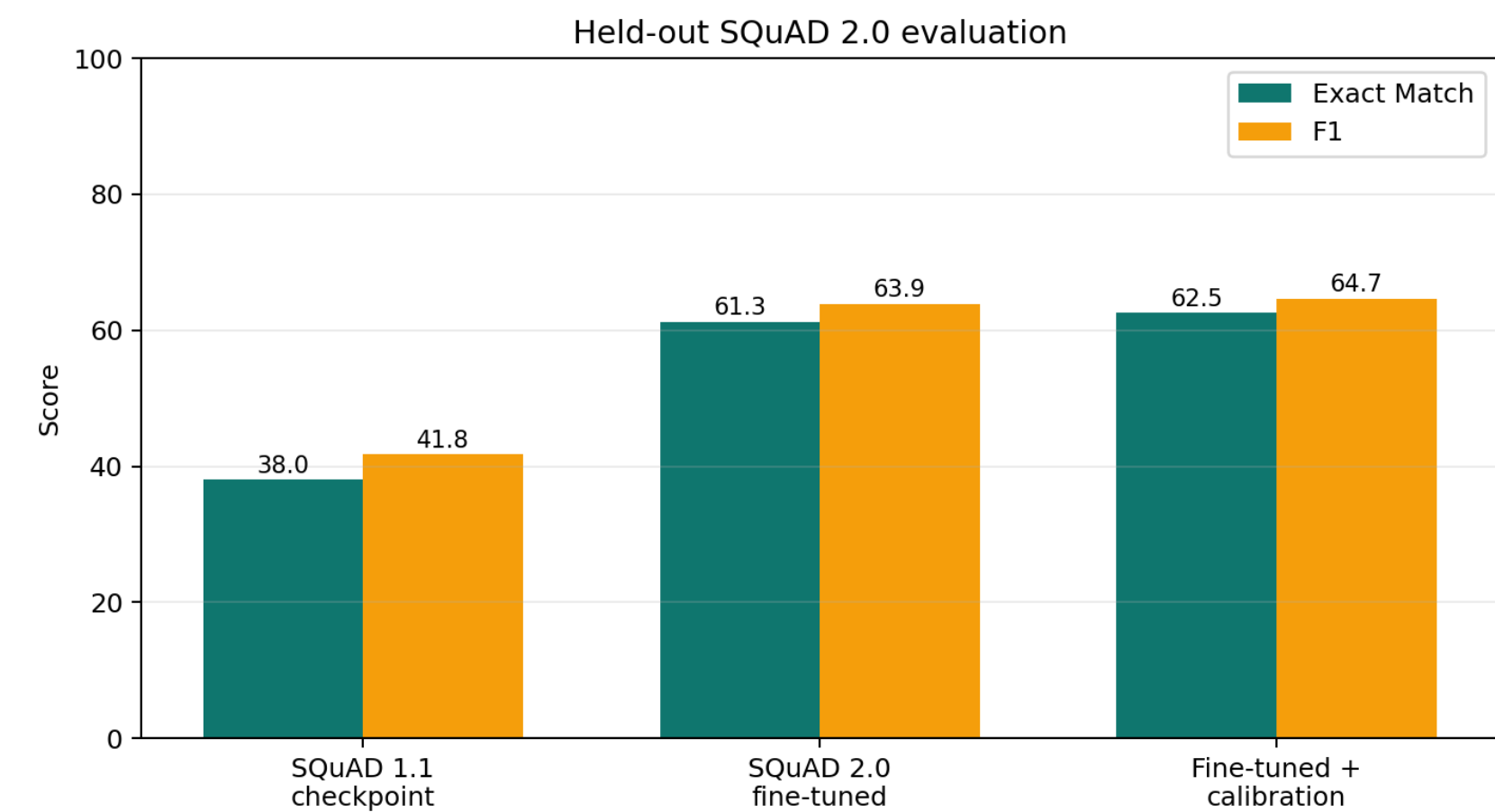
Fine-tuning assigns both gold positions to [CLS] for unanswerable examples and windows without the answer. Calibration selects τ on a disjoint split, freezes it, then evaluates once on held-out data.



Reproducibility

Seeded split, unit tests, smoke run, exact configuration, model checkpoint, tokenizer, software versions, raw metrics, predictions, figures, and 20 auditable error cases are included. The code package excludes data and model weights.

Results



Fine-tuning gain: +23.27 EM / +22.12 F1

Calibration gain: +1.27 EM / +0.76 F1

Final: 62.53 EM / 64.67 F1 / 70.94% no-answer accuracy

Error analysis & conclusion

Corrected: unsupported entities and reversed premises. Example: a passage calls the Ilkhanate an ally, while the adversarial question asks for a Persian enemy; the final system abstains.

Remaining: span-boundary mismatches ('About 61.1%' vs. '61.1%'), long-context misses, and over-abstention.

Takeaway: Answer extraction and the decision to answer are distinct capabilities. SQuAD 2.0 adaptation supplies the main improvement; held-out calibration cheaply adjusts the reliability/recall trade-off.